

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458316.8725 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.215 +/- 0.063
Transit depth (Rp/Rs)^2	4.63 +/- 2.69 [%]
Orbital Inclination (inc)	89.12 +/- 2.82
Airmass coefficient 1 (a1)	0.35 +/- 0.17
Airmass coefficient 2 (a2)	0.61 +/- 0.22
Scatter in the residuals of the lightcurve fit is	40.67 %
Best Comparison Star	None
Optimal Aperture	2.29
Optimal Annulus	3.25
Transit Duration (day)	0.0878 +/- 0.0091

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.215 ± 0.063 vs 0.1646 (deviation 0.69σ)
Duration fit vs published	0.0878 d vs 0.0758 d (deviation 1.14σ)
Mid-time O–C	9.2 min
Depth SNR	3.0
Residual scatter	40.67 %

Light curve

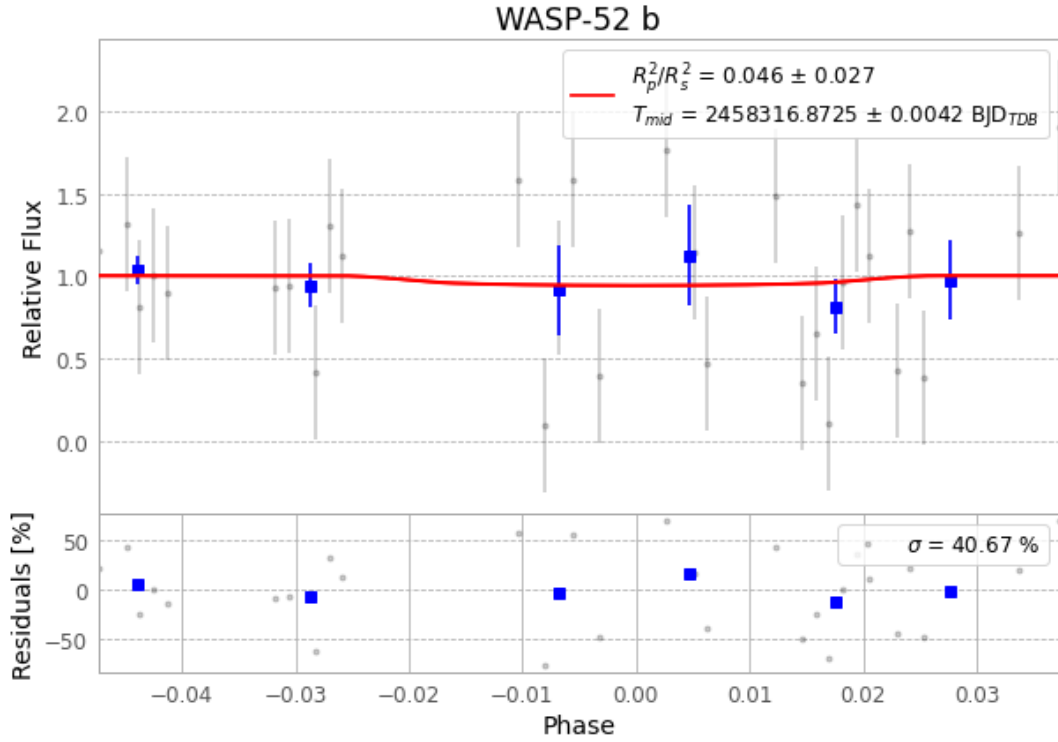


Figure 1 — FinalLightCurve_WASP-52 b_2018-07-16.png (SHA-256 5e4c008395f891d0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [248, 423], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ec63b3db8f67a336...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a2b4e71

Data provenance

77 FITS frames (51 MB), WASP-52180717065112.FITS → WASP-52180717103912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-180717.log (e588c4b88206...), FinalLightCurve_WASP-52 b_2018-07-16.png (5e4c008395f8...), FinalLightCurve_WASP-52 b_2018-07-16.csv (8fb8d6149be8...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-21 01:24Z.